CS771 Mini-Project 1: Next-Day PM2.5 Forecasting and Heatmaps

Group Members:

Chirag Gupta (240311)
Lakshya Arukiya (240589)
Arpit Kumar Singh (240194)
Priyanshu Verma (240811)
Saurabh Raj (240947)

PM2.5 is in μg/m^3 throughout.

Part 0: Cleaning the raw Kanpur data

Format. 76 sensors on a 15-minute UTC grid (1 Jul – 31 Aug 2026), three ID styles:

- **44 ten-digit IDs:** whole-number values; mean 42.1, median 37; occasional 999 codes.
- **31 eight-character IDs:** decimal values; mean 37.3, median 33.0.
- **One 12-character ID (AC0BFBE34CC7):** whole numbers; mean 24.4.

All, except AC0BFBE34CC7 report hourly (median gap 60 min). It reports every 15 min. Sensors report in 51–88% of the 1,488 hours (median 80% for ten-digit, 68% for eight-character). No sensor moves: each one's coordinates are constant.

Hourly table. Readings are averaged within each UTC hour, labelled by the hour's start (82,131 sensor-hours). Only AC0BFBE34CC7 has more than one reading per hour. For it, the hourly mean differs from the :00 reading by 2.6 on average (max 18.5, RMSE 3.5). The mean uses all four readings, so it is less noisy.

Flags (nothing deleted, no gaps filled):

Flag	Rule and justification	Readings	Sensors
error_code	value = 999 (or negative): a device code, far above any plausible ambient level	136	5 (1204260025 has 120)
zero	value = 0: ambient PM2.5 is never exactly 0 in Kanpur; only 1.3% of good readings are below 5	49	19
stuck	same value for $\geq$ 24 consecutive reported hours: real air changes over a day	559	5
dropped_sensor	$>$ 50% of the sensor's readings flagged	1,734	2 (1203260068, 1203260082)
ok	none of the above	79,653	74

Outages: no hour has fewer than 10% of sensors reporting.

Sensors that barely follow the network: correlation of each sensor's good readings with the network median is lowest for 1204260025 (r = 0.18), U8APQDBX (0.26) and 1203260075 (0.29).

Calibration.

- **Pairs:** 23 ten-digit sensors have an eight-character sensor within 3 km.
- **July fit:** 8-char $\approx 0.21 \times$ 10-digit + 29.7.
- **Tested on August:** MAE rose from 12.33 to 15.27 (+24%); RMSE fell slightly, from 23.34 to 23.00 ($-$1.5%).

The correction does not help. Paired readings agree poorly (July r = 0.25): sensors up to 3 km apart measure different air, and the ten-digit family has spikes. With a noisy predictor, least squares shrinks the slope towards 0 (regression dilution). The "correction" then pulls every reading towards the eight-character mean. That trims a few large squared differences, so RMSE drops slightly, but it adds error on typical hours, so MAE rises. The family means are almost equal (36.2 vs 37.3), so there is little constant bias to remove. Without co-located sensors or a reference monitor, device differences cannot be separated from real spatial differences.

Stage	Sensors	Readings	Share missing
Raw (15-min grid)	76	85,762	81.0%
Hourly table	76	82,131	27.4%
Clean (flag = ok)	74	79,653	27.7%

The 81% on the raw grid is mostly because most sensors report hourly, not lost data. The daily cycle (IST) peaks at 06:00 (44) and 19:00 (43), with a minimum around 15:00 (35): morning and evening emissions into a shallow boundary layer, then afternoon mixing.

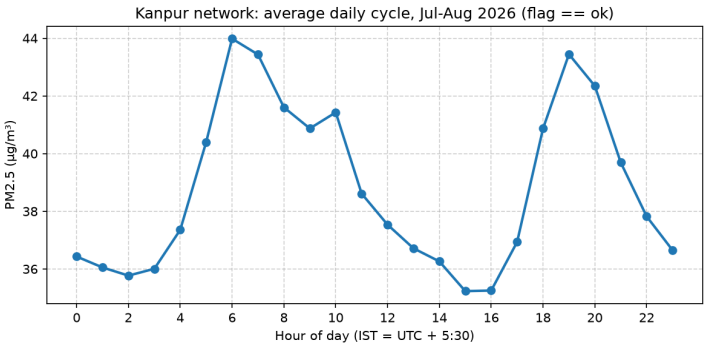


Fig. 1: Kanpur network mean by hour of day (IST), flag = ok.

Part 1: Baselines

Setup.

- **Issue and target:** issue at 12:00 on day d; target 12:00 on day $d+1$.
- **Training:** every target from 3 Jan to 31 Aug that has 24 h of history (242 days $\times$ 468 stations = 113,256 rows).
- **Validation:** every station and every target from 1 Sep to 30 Nov (91 $\times$ 468 = 42,588).
- **Persistence:** today's 12:00 reading.
- **Ridge:** the 29 required features, `Ridge(alpha=1)` on unscaled features, all stations pooled, negative forecasts clipped to 0.

Every feature uses readings at or before the issue time.

Month	Persist. RMSE	Ridge RMSE	Persist. MAE	Ridge MAE
Sep	23.33	**20.34**	8.91	10.35
Oct	20.22	**19.36**	11.97	13.46
Nov	47.81	**43.06**	31.70	**30.81**
Overall	32.75	**29.58**	**17.47**	18.16

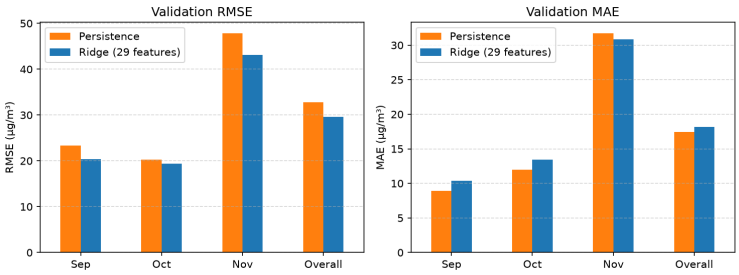


Fig. 2: Validation RMSE and MAE per month.

Does one baseline win on both metrics? Why can a squared-error model lose on MAE?

No. Ridge wins on RMSE; persistence wins on MAE overall and in September and October.

- **Why Ridge wins RMSE:** squared error is dominated by the few large errors, and Ridge is fitted to reduce exactly those. Errors above 100 fall from 2.09% to 1.51% of forecasts, and the 99th-percentile error from 136.8 to 115.9.
- **Why it loses MAE:** it shrinks towards the pooled training level. Its 24 lag weights sum to only 0.72, and the training months include the very polluted January (noon mean $\approx$ 169). So it over-forecasts the cleaner autumn days: mean error +5.3 vs -0.5 for persistence, median absolute error 11.3 vs 8.8.

In short, a squared-error model estimates the conditional mean and gives up many small errors to reduce a few large ones. MAE rewards getting the typical (median) day right.

Part 2: Improved forecast

Method. The final model is **0.5 $\times$ Ridge(alpha = 100) + 0.5 $\times$ Random Forest** on 35 features. Forest settings: 100 trees, min_samples_leaf = 20, max_features = 0.33, max_samples = 0.5, seed 42. Training uses the same 113,256 rows as Part 1. Alpha, the forest settings and the 0.5 weight were chosen on validation (weights from 0.4 to 0.6 give almost identical results).

Feature selection. We first fitted Ridge on 7 candidate groups (42 features) and removed one group at a time:

- **Kept:** network_context (+0.97), last_24h_readings (+0.43), temporal_calendar (+0.10) and spatial_neighbourhood (+0.09).
- **Dropped:** short_term_trends (0.00), lag_same_hour at $t-48/-72$ h (-0.03) and station_training_mean (-0.06).

Month	Persistence RMSE / MAE	Ridge-29 RMSE / MAE	Ridge-35 RMSE / MAE	Forest RMSE / MAE	Final blend RMSE / MAE
Sep	23.33 / 8.91	20.34 / 10.35	20.25 / 10.22	20.31 / 7.92	**19.98** / **8.69**
Oct	20.22 / 11.97	19.36 / 13.46	18.92 / 13.14	18.36 / 11.75	**17.85** / **11.68**
Nov	47.81 / 31.70	43.06 / 30.81	42.45 / 29.97	43.95 / 30.64	**42.42** / **29.71**
Overall	32.75 / 17.47	29.58 / 18.16	29.17 / 17.72	29.79 / 16.72	**28.87** / **16.64**

The blend beats both baselines on both metrics, overall and in every month. Ridge on the same 35 features alone beats both on RMSE but loses to persistence on MAE (17.72 vs 17.47), which is why the forest was added.

Ablation from the final model (remove one group, refit both members with the same settings; full model RMSE 28.87, MAE 16.64):

Group removed	Features	Val RMSE	ΔRMSE	ΔMAE
last_24h_readings	24	36.22	+7.35	+5.05
network_context (mean at t, 24 h mean, std at t)	3	29.27	+0.40	+0.89
temporal_calendar (year and weekday sin/cos)	4	29.17	+0.30	-0.19
spatial_neighbourhood (10 and 25 nearest, at t and 24 h)	4	29.00	+0.13	+0.17

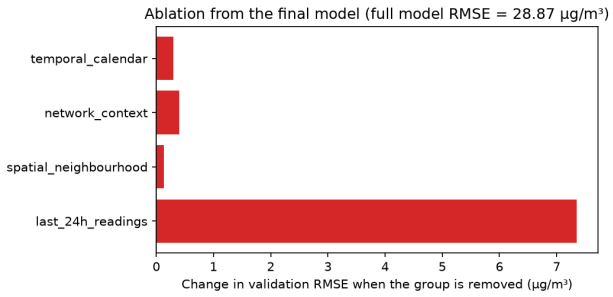


Fig. 3: Change in validation RMSE when each group is removed from the final model.

What did each group add? What did not help?

- **The station's own last 24 h** carries most of the skill.
- **Network context** adds the state-wide level and spread, which capture regional episodes.
- **The calendar terms** help on large November errors (RMSE) but slightly hurt typical days (MAE).

- **Neighbour means** add little because they overlap with the other groups. The network mean already gives the regional level, and the data were gap-filled from neighbours, so each station's own series already contains neighbour information.
- **Did not help:**
  - *Trends* are exact linear combinations of lags already in the model.
  - *Same-hour readings 2–3 days back* add noise beyond the last 24 h.
  - *The station training mean* mixes the polluted winter with the clean monsoon, so it says little about autumn.

Did a more flexible model beat Ridge?

Not on its own:

Model	Train RMSE	Val RMSE
Ridge	27.89	29.17
Random Forest	21.15	29.79
HistGradientBoosting	20.55	31.28

Both tree models fit the training months much better but do worse on validation: they overfit relationships specific to the winter and monsoon regimes, which do not carry over to the Sep–Nov build-up. This is not a range problem: training targets reach 500, the January mean (169) is above November's (104), and no validation target exceeds the training maximum. The forest still has the best single-model MAE (16.72), so it gets typical days right and misses large November swings. Ridge does the opposite. Their errors are only partly correlated, so averaging them beats both.

What missing information would help most?

Meteorology (boundary-layer height, wind, humidity, rain), crop-fire activity in Oct–Nov, a festival/emissions calendar (Diwali was 31 Oct – 1 Nov 2024), and flags marking which Bihar values were gap-filled.

Part 3: Lead time

The target stays at 12:00 on day $d{+}1$; only the issue time moves. Both learned models are refitted for each lead, with their features taken relative to the new issue time. RMSE (MAE):

Lead	Issue time	Persistence	Ridge-29	Final blend
1 h (nowcast)	11:00, $d{+}1$	23.95 (10.36)	18.95 (8.67)	**18.60 (8.16)**
3 h	09:00, $d{+}1$	51.43 (28.28)	28.51 (16.70)	**27.15 (14.88)**
6 h	06:00, $d{+}1$	66.82 (41.28)	34.06 (23.16)	**30.22 (18.08)**
12 h	00:00, $d{+}1$	72.74 (46.74)	31.97 (21.31)	**29.34 (17.56)**
24 h (day-ahead)	12:00, d	32.75 (17.47)	29.58 (18.16)	**28.87 (16.64)**

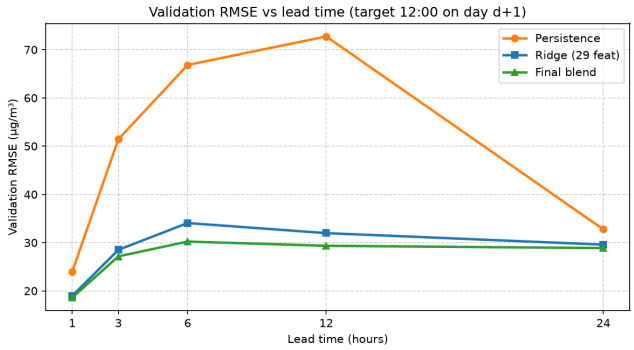


Fig. 4: Validation RMSE vs lead time.

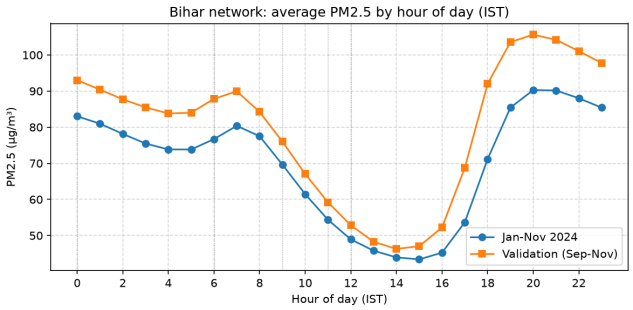


Fig. 5: Bihar network mean PM2.5 by hour of day (IST).

How does persistence behave across leads, and why?

Its error grows from 1 h to 12 h, then drops sharply at 24 h. The reason is the daily cycle (Fig. 5). Over the validation period the network mean is about 93 at 00:00, 88 at 06:00, 76 at 09:00 and 59 at 11:00, but only 53 at 12:00. Persistence from a night or morning reading therefore carries a systematic over-forecast of roughly 25–40 into the noon target. At 24 h the issue hour equals the target hour, so the mismatch disappears and only the day-to-day change remains.

Do the learned models behave the same way?

No. Because they are refitted per lead, they learn the offset between the issue hour and noon, and show no large dip. Their curve is still not monotone: a night-time reading predicts noon worse than yesterday's noon reading does, so issuing at 00:00 or 06:00 is no better than issuing at 12:00 the day before.

Which lead time for a public warning service?

The 24 h forecast issued at 12:00 on day d. It gives a full day to act. Compared with a 1 h nowcast it costs about 10.3 in RMSE (28.87 vs 18.60) and 8.5 in MAE (16.64 vs 8.16). Waiting until 00:00 or 06:00 would gain nothing (29.34 and 30.22).

Part 4: Forecast maps vs observed maps

Method. Station values are interpolated to the 3,396-point 0.05° grid by inverse-distance weighting (power 2, all 468 stations). The same weights are used for observed, forecast and persistence values. Days are target days at 12:00 IST:

- **Clean day:** lowest network mean, 15 Sep (7.9).
- **Polluted day:** highest network mean, 27 Nov (175.0).
- **Worst day for our model:** highest station RMSE, 29 Nov (82.2; persistence 112.0).

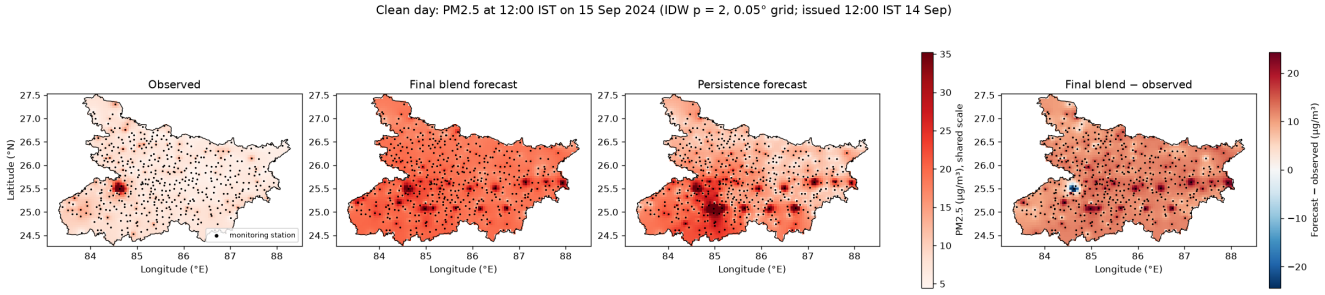


Fig. 6: Clean day, 15 Sep 2024 (station RMSE: model 18.2, persistence 27.3).

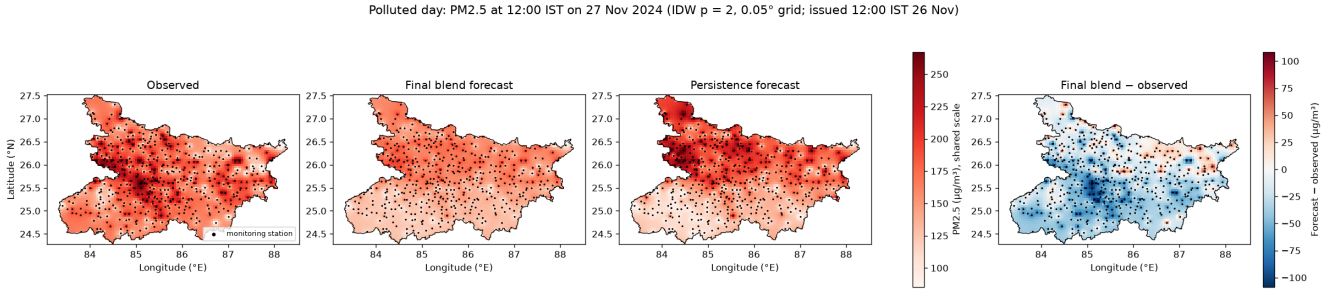


Fig. 7: Polluted day, 27 Nov 2024 (station RMSE: model 67.6, persistence 71.7).

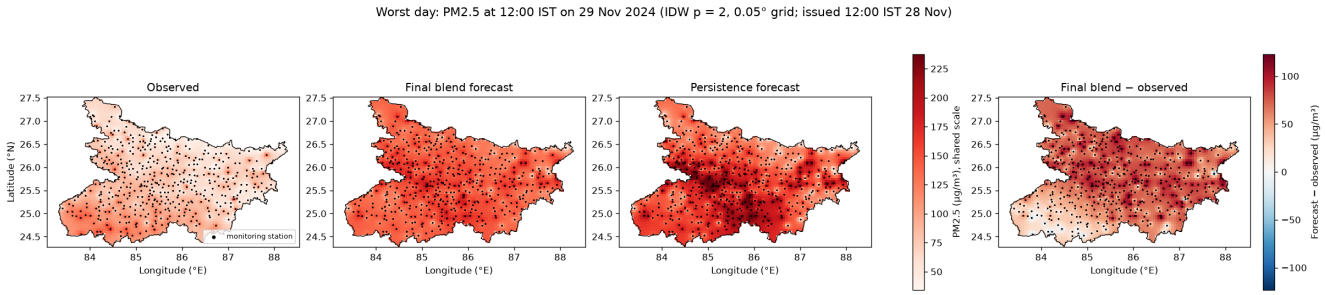


Fig. 8: Worst day for the model, 29 Nov 2024. The model expected yesterday's haze to persist, but it cleared. In Figs. 6–8 the three concentration panels share one colour scale; the error panel has its own symmetric scale.

Daily averages over the 91 validation days:

	Station RMSE	Map RMSE	Pattern correlation
Persistence	28.57	15.37	0.580
Final blend	**25.24**	**14.35**	**0.600**

Station RMSE here is the mean of the daily RMSEs; the pooled values are 28.87 and 32.75.

Which RMSE is better, and which should we quote?

Map RMSE is far lower because each grid value is a weighted average of many stations, so errors that point in different directions cancel. The observed map is smoothed in the same way, so both lose the station-level detail that makes forecasting hard. Station RMSE should be quoted, because it is the only one computed against real measurements.

Why is the observed map not ground truth?

It is itself a model output: a smooth weighted average between stations. It cannot show local hotspots or sharp gradients, and it inherits sensor errors and the upstream gap-filling. Comparing against it measures agreement with an interpolation, not with the true PM2.5 field.

When do pattern correlation and RMSE disagree?

A forecast with the right spatial pattern but 50 too low everywhere has a correlation near 1 and an RMSE near 50. Fig. 7 shows a mild version: the right broad shape, under-forecast across most of the state. For public warnings RMSE (the absolute level) matters more, because advisories are triggered by absolute thresholds.

Part 5: Hidden December test

The final model is refitted on every target from 3 Jan to 30 Nov (333 days $\times$ 468 = 155,844 rows) and saved as `model.pkl` (12.3 MB; holds both members).

- **What `forecast.py` does:** loads the model once at import, builds the same 35 features from `history` only (readings at or before the issue time), and never refits.
- **Consistency check:** the notebook confirms that `forecast.py` reproduces its predictions exactly.
- **Official checker:** format OK; 0.03 s median and 0.06 s maximum per call.
- **November sanity check:** RMSE 36.18 and MAE 24.87, vs persistence 47.81 and 31.70. November is inside the refit data, so this is not a test score.

Limitations

- **Gap-filling:** the Bihar values were gap-filled upstream without marking which ones, which smooths stations towards their neighbours.
- **Single validation period:** all choices were made on one period (Sep–Nov), which is a seasonal transition.
- **Large-day errors:** November errors remain large (RMSE $\approx$ 42) because sudden episodes are driven by weather and fires we cannot observe.
- **December shift:** December may differ from all validation months.
- **Kanpur calibration:** the pairing only tests sensors up to 3 km apart, not co-located ones.

Use of AI tools

- **Claude Code (Anthropic, Claude Opus 5.5):**
  - A reference implementation of the Part 1 baselines and the RMSE-vs-MAE diagnostics.
  - A review of the submission for leakage and for mismatches between code and report.
  - The code that refits and saves `model.pkl`, generates every figure, computes the lead-time curve for all methods and runs the ablation from the final model.
  - Proposing and testing the Ridge + Random Forest blend.
  - Drafting this report's text.

All AI-produced code was run end to end in the notebook, and every number above is a notebook output. We checked this work and take responsibility for it; every member can explain any part in the viva.